Coding / Protocol

PDF edition of preserved repository document | docs/coding/protocol.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Coding protocol v1

The flagship implements bounded Python repair with actual edits to disposable repositories and actual function execution in capability-isolated CPython WASI. The host compares private expected values. Model-visible observations contain visible tests only. Hidden evaluation occurs after the policy stops. Goal verification cannot be satisfied by a model's success declaration.

The selected model is Qwen2.5-Coder-1.5B-Instruct, revision 2e1fd397ee46e1388853d2af2c993145b0f1098a, Apache 2.0. A locked ten-task validation pilot produced 9/10 actual repairs, versus 7/10 for the 0.5B model. This small selection pilot is not a general capability estimate. Full model weights remain outside Git in the local cache.

Four policies share this backbone: model-only source repair, structured state repair, trajectory-example repair, and bounded experiential repair. The last policy sees actual visible-test feedback and has up to three sequential attempts. Its success@3 is an interactive recovery measure, not HumanEval pass@3. Its memory uses executed teacher examples. The first two controls do not acquire memory as the sample budget changes, so their frozen results are reused transparently rather than counted as new independent seeds.

The first matrix measures retrieval adaptation at nine sample budgets (10 through 5000) and five independently shuffled training-memory seeds. These are retrieval budgets, not gradient-training budgets; gradient DER is not identifiable from this matrix. Teacher patches and trajectories constitute more supervision than the controls receive. Extra tool execution and retries constitute more inference compute. Any difference is conditional on those information and compute differences.

Task families are intentionally simple synthetic Python functions. IID splits share structural families but distinct names, parameters, and repository hashes. OOD families are withheld from training: function composition, cross-file dependency, compound repair, and filtering structure. These four families do not represent broad repository engineering. Exact-hash nonoverlap cannot establish semantic independence for shared IID templates. Public benchmark pretraining contamination is unknown.

WASI preopens only curated standard-library files and the current disposable task repository, both read-only to generated code. Host tools apply validated edits. Runtime probes deny unrelated file access, sockets, and subprocess execution. Fuel, memory, CPU, output-file size, and host wall deadlines bound execution. Native macOS sandbox creation failed with Operation not permitted; Docker and Podman are absent. Generated code never executes in host Python.

The pinned official HumanEval revision is 6d43fb980f9fee3c892a914eda09951f772ad10d (164 tasks). Independent greedy completions receive no hidden-test feedback. CPython WASI differs from the original host runtime; report incompatibilities as failures and publish the runtime. With one completion per task, only pass@1 is available. No pass@k estimate is produced for k greater than one.

train_lora.py provides separate real LoRA gradient adaptation (rank 8, query/value projections, masked prompt loss, AdamW, one epoch). Its resulting records and checkpoints must not be conflated with the retrieval matrix. No gradient threshold claim is permitted without evaluating checkpoints at matched training budgets.

Exact prompt/model/revision/token-limit repeats under greedy sampling may reuse an on-disk completion cache. Cached calls are recorded with zero new model calls/tokens and zero generation time, plus the original measured timing. Actual repository execution is repeated for each evaluated policy. Cache hits are not additional independent model samples; latency summaries must distinguish fresh calls from reuse. Changing the model revision or any prompt byte changes the cache key.

The earlier 1000-task/three-seed configuration was a preflight only: no held-out model evaluations were run. Its locked validation/test/OOD tasks are retained during expansion to 5000 training tasks; all held-out seeds are excluded from the additional training pool. Executed prefix teacher records are reused with their actual provenance.

The independent LoRA diagnostic exposed a source-envelope mismatch: correct-looking JSON repository maps were rejected by the initial path/content-only parser. A general normalization now accepts a single repository-map edit, a typed edit, path/content, or a source block whose target is unambiguously identified by the public entry point. Source text is never repaired by the parser. Ambiguous and escaping paths remain rejected. The original held-out sets are therefore diagnostic, retained in full; a fresh primary set (seed 9001, 20 IID/20 structural OOD tasks) was frozen before primary comparisons. Training remains the same 5000-task pool, and fresh held-out seeds exclude all training and prior evaluation seeds. This disclosed protocol revision prevents diagnostic exposure from masquerading as untouched testing.

Full typed expert tuples are archived separately from the compact model-facing examples. The accepted preflight prefix is reexecuted to collect genuine state/action/event/result/next-state records; archived results are never synthesized from labels. A teacher repair that fails actual visible execution stops collection. These provenance additions do not change model prompts or proposed source edits.

Completion adds a bounded two-branch static planner. Each proposed action and alternative produces a hypothetical state, a hypothetical follow-up test action, and a heuristic precondition/syntax score. This is not a learned world model and never predicts authoritative test success. Planning can reject an invalid predicted edit in favor of a real observation action; the no-dream ablation measures its actual effect. Stage timers retain full durations in transcripts; irrelevant timing metadata is excluded from model prompts for deterministic prompt reuse. Previously measured adapter results and old protocol records are retained as their original controller versions.

Frozen completion comparison

The preceding sections preserve the historical retrieval/diagnostic protocol. The completion comparison uses pinned Qwen2.5-Coder-1.5B LoRA checkpoints, gradient budgets 10/25/50/100, learning seeds 11/23/37 and 20 IID plus 20 OOD tasks per cell. The independent lockbox uses seed13001, 50 IID plus 50 structural OOD tasks, checkpoint budget100 and training seed11 in each of A/structured/B/C. A and structured produce one patch; B/C retain up to eight proposals and three accepted edits. The exact frozen source/checkpoint/task hashes are in emergency_mps_v1/locked_protocol.json.

The operational continuation uses frozen batch16/float16/MPS, keeps model/tokenizer resident and disables generation caching. Original CPU float32 learning evidence remains unchanged and separately identified. The original emergency deadline is historical metadata; the later completion request removes that operational cutoff only. No new scoring, held-out tuning, hidden feedback or altered training data is introduced. Proposal

rejection is separate from accepted-transition validity; actual repair success is independently assessed at terminal execution.